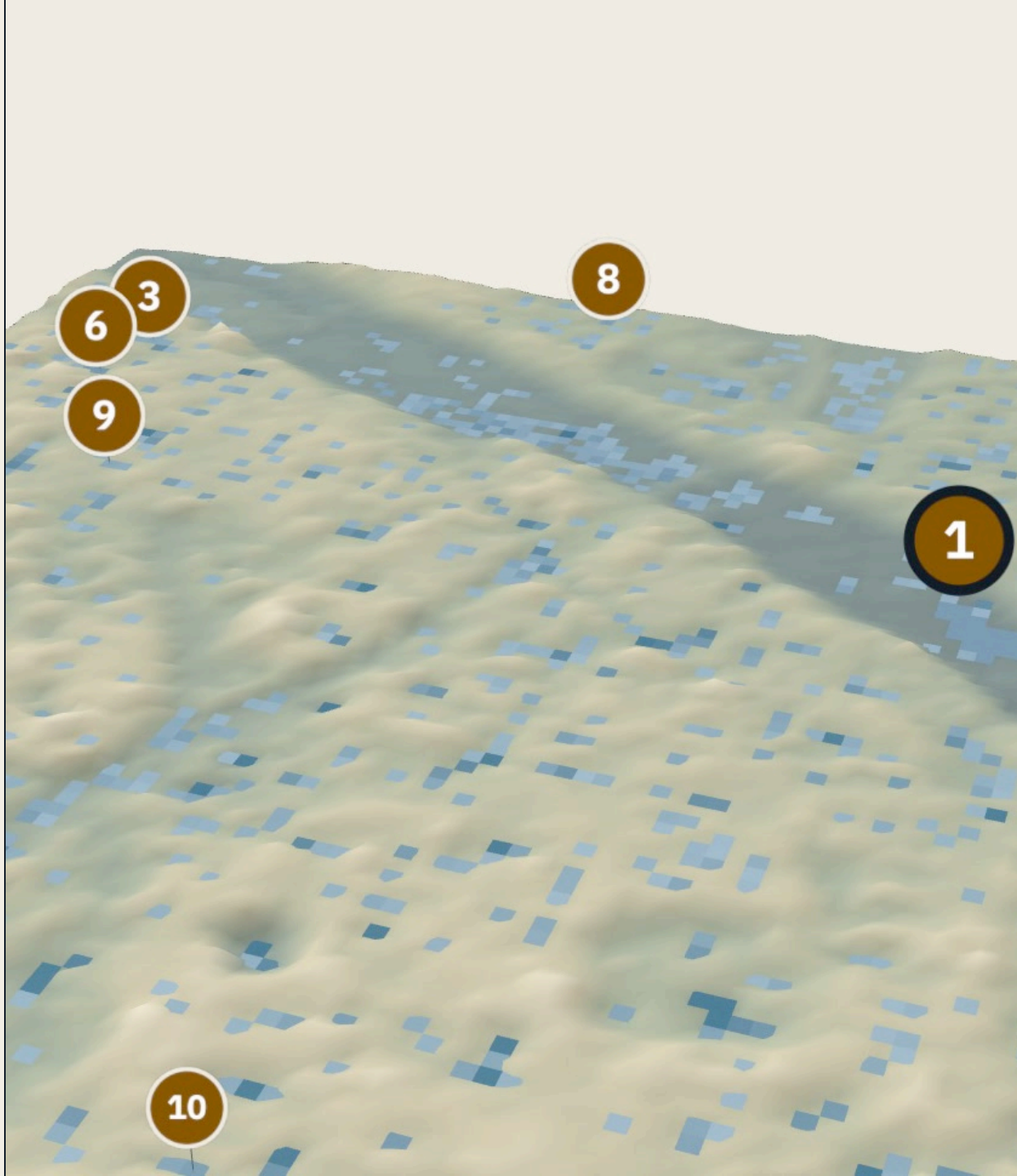
PaaniPulse

See where rainwater goes.

India-wide terrain and rainfall scenario explorer, running entirely in the browser.

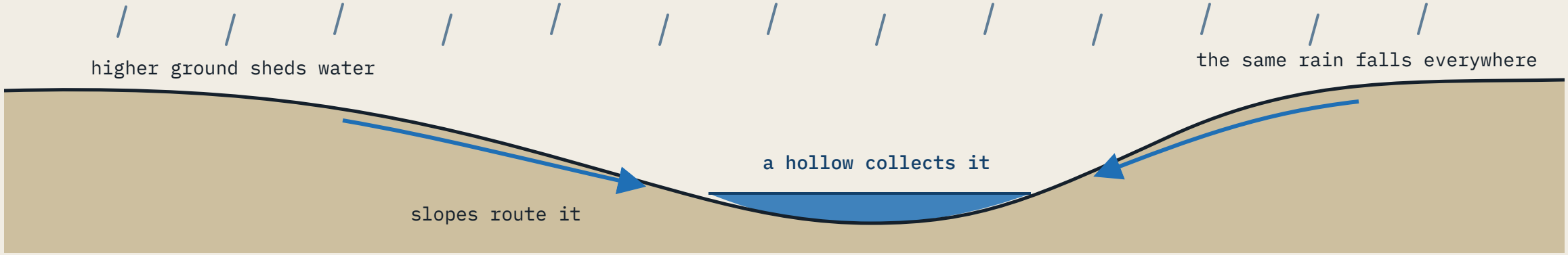
HackCult '26 Blithchron '27 · IIT Gandhinagar



THE PROBLEM

Rainfall says how much water falls. Terrain decides where it goes.

Every monsoon, Indian cities take heavy rain. The same rainfall depth can leave one street dry and another under water, because the ground directs the flow.



01

Terrain controls movement

Water runs downhill along the steepest path available.

02

Elevation shapes drainage

Small height differences decide which way an area drains.

03

Depressions fill first

Low spots with no outlet are where water can gather.

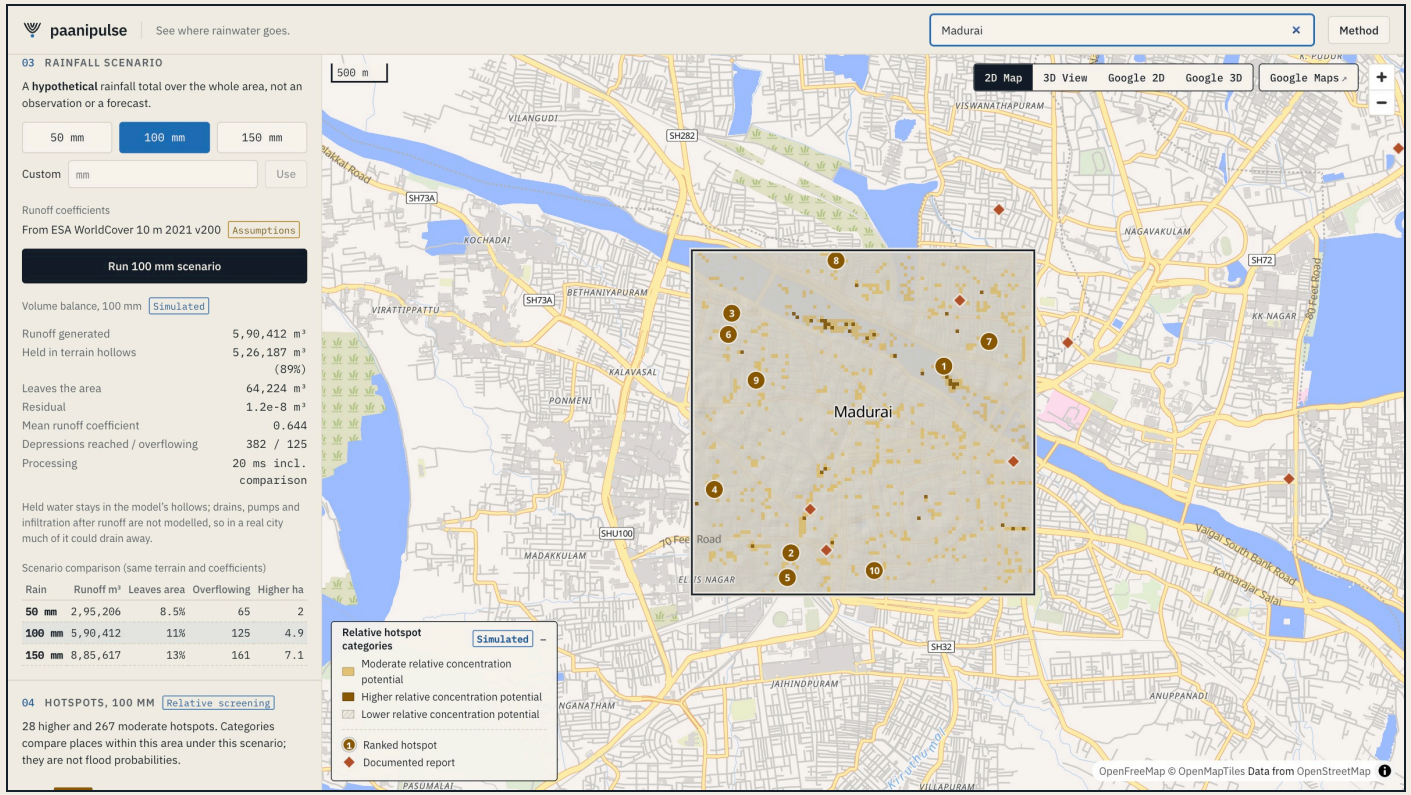
04

Rain data alone is incomplete

Without terrain context, a rainfall number cannot show where water may collect.

THE SOLUTION

PaaniPulse puts terrain and rainfall in one map.



Live app · Madurai, 3 × 3 km, hypothetical 100 mm scenario, hotspot layer

01 **Explore India.** Search 1,618 bundled places, or any place via OpenStreetMap search.

02 **Load real elevation.** Copernicus DEM GLO-30, about 30 m per cell.

03 **Trace water.** Filled hollows, flow direction, accumulation and HAND layers.

04 **Try a rainfall.** 50, 100 or 150 mm, or a custom total.

05 **Find hotspots.** Ranked places where the model concentrates water.

06 **Understand why.** Plain-language reasons built from each place's own values.

HOW IT WORKS

From a place name to an explained result.

1 Select a location

Search a place or draw an area up to 25 km².

`bundled gazetteer · Photon · Nominatim`

2 Load elevation and land cover

Only the bytes covering the area are read from cloud-optimised GeoTIFFs.

`Copernicus GLO-30 · ESA WorldCover 2021`

3 Analyse the terrain

Fill hollows, route flow cell to cell, accumulate upslope area, measure HAND.

`Web Worker hydrology engine`

4 Choose a rainfall scenario

A hypothetical total spread evenly over the area.

`50 · 100 · 150 mm or custom`

5 Estimate runoff and accumulation

Runoff by land cover, then fill-and-spill through the hollows with a full volume balance.

`rational-method coefficients`

6 Explore and understand

2D map, 3D terrain, ranked hotspots and a plain-language explanation.

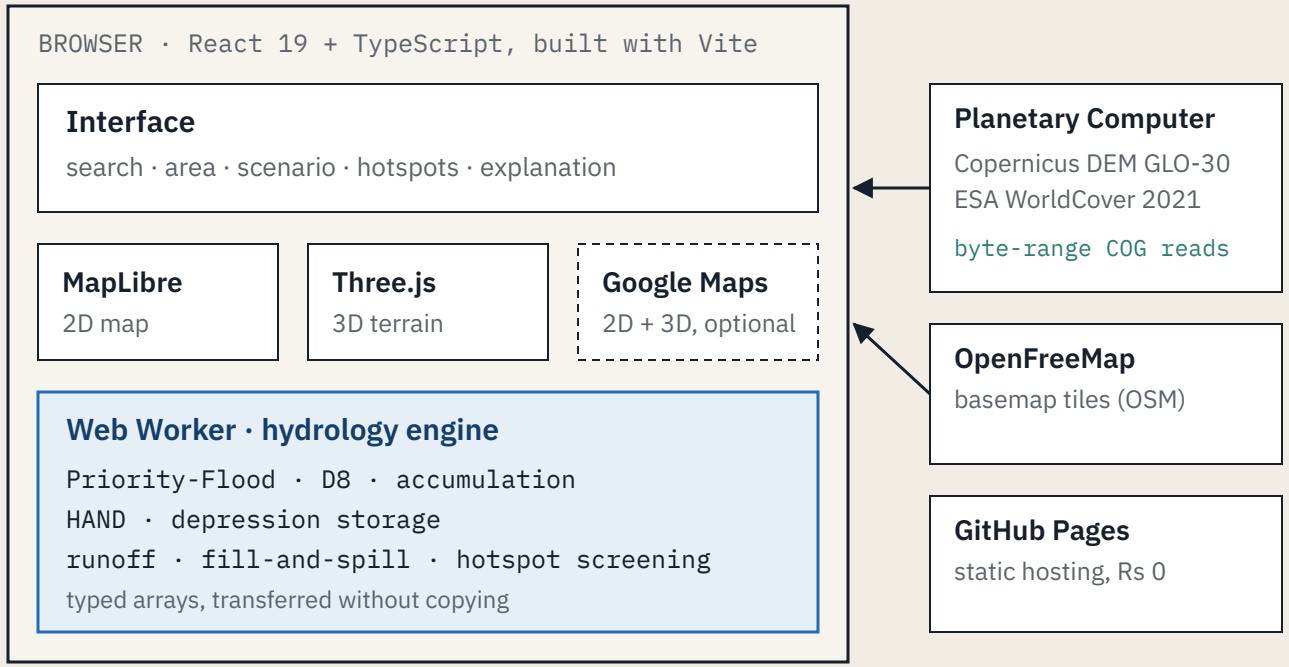
`MapLibre · Three.js`

Everything runs in the visitor's browser: no backend, no database, static hosting on GitHub Pages.

`3 × 3 km Madurai analysis: under 20 ms`

THE TECHNOLOGY

A hydrology engine that runs in the browser.



Priority-Flood depression filling
Raises every enclosed hollow to its spill level and keeps the depth as storage.

D8 flow direction and accumulation
Each cell drains to its steepest neighbour; summing gives the upslope area.

Height Above Nearest Drainage (HAND)
How high a cell sits above the channel it drains to.

Land-cover runoff
Rain × a documented coefficient per WorldCover class, editable in the app.

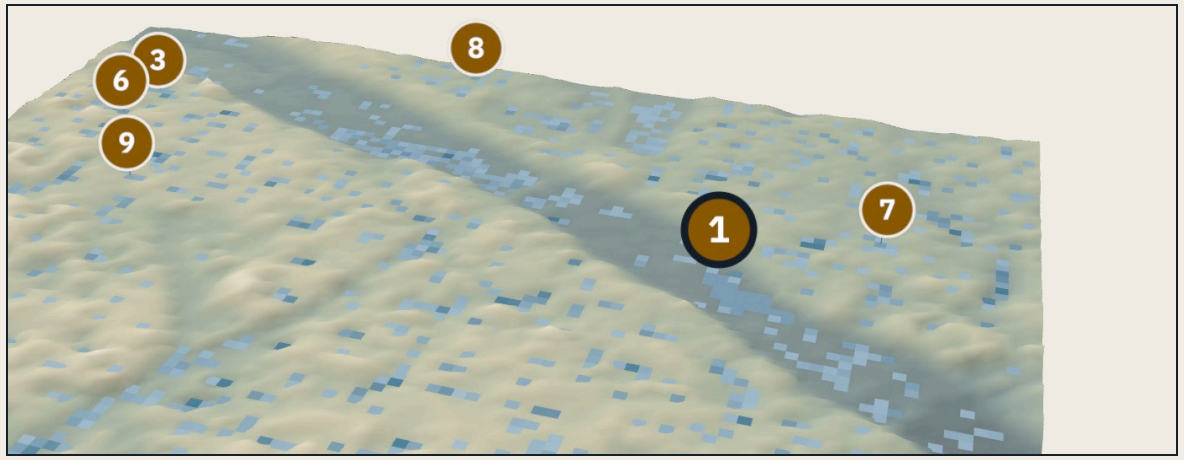
Hypothetical rainfall scenarios
Runoff fills hollows and spills downstream; a 7-point screen ranks hotspots.

Data model
No database: typed arrays in the browser, one value per elevation cell.

ElevationGrid	HydroPayload	ScenarioPayload	Hotspot
width, height, bounds, crs	directions, accumulation	rainfallMm, runoffM3	rank, category, peak
values: Float32Array (m)	hand, fillDepthM, depressions	pondDepthM, summary	criteria, flags, confidence

PRODUCT EXPERIENCE

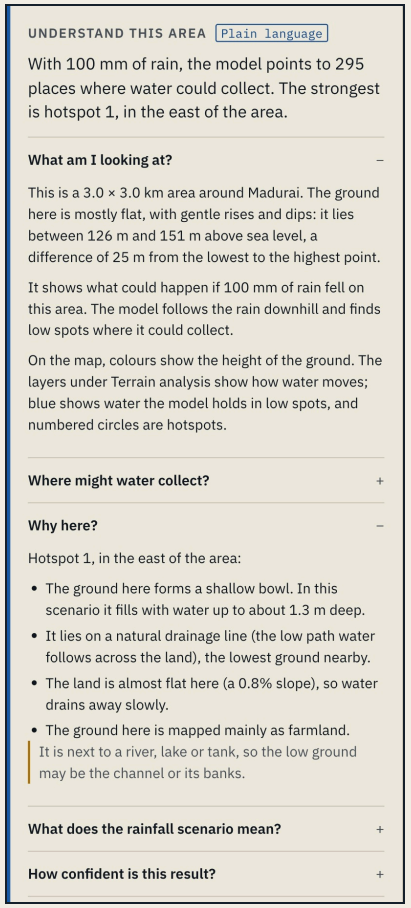
One area, every view, plain words.



3D View (Three.js): real GLO-30 heights ×3, modelled ponding, hotspot pins



Google 3D: PaaniPulse ponding and hotspots, with rain · imagery © Google



UNDERSTAND THIS AREA Plain language

With 100 mm of rain, the model points to 295 places where water could collect. The strongest is hotspot 1, in the east of the area.

What am I looking at? –

This is a 3.0 × 3.0 km area around Madurai. The ground here is mostly flat, with gentle rises and dips: it lies between 126 m and 151 m above sea level, a difference of 25 m from the lowest to the highest point.

It shows what could happen if 100 mm of rain fell on this area. The model follows the rain downhill and finds low spots where it could collect.

On the map, colours show the height of the ground. The layers under Terrain analysis show how water moves; blue shows water the model holds in low spots, and numbered circles are hotspots.

Where might water collect? +

Why here? –

Hotspot 1, in the east of the area:

- The ground here forms a shallow bowl. In this scenario it fills with water up to about 1.3 m deep.
- It lies on a natural drainage line (the low path water follows across the land), the lowest ground nearby.
- The land is almost flat here (a 0.8% slope), so water drains away slowly.
- The ground here is mapped mainly as farmland.

> It is next to a river, lake or tank, so the low ground may be the channel or its banks.

What does the rainfall scenario mean? +

How confident is this result? +

Understand this area

India-wide exploration
Any place, any area up to 25 km².

2D, 3D and Google 3D
Switch views without losing the area or scenario.

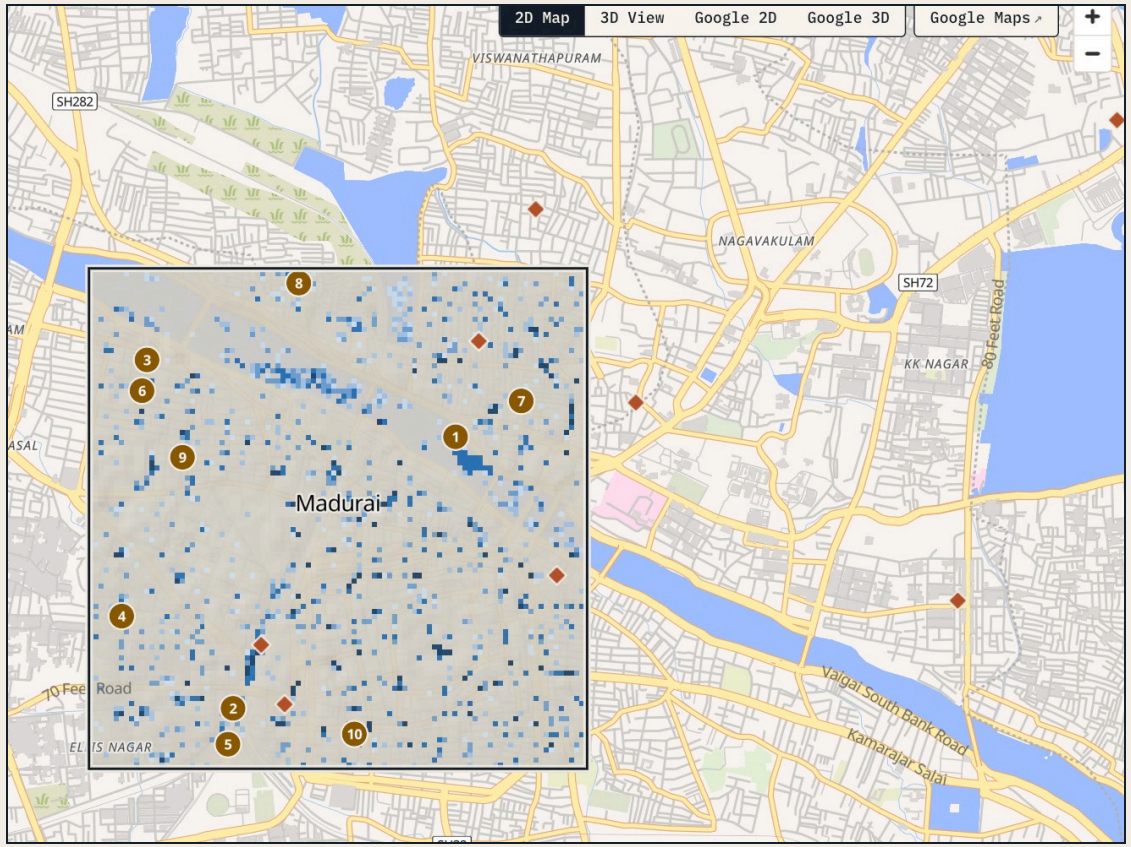
Rainfall controls
50, 100, 150 mm or custom, with a full volume balance.

Hotspots
Top 10 numbered on the map, each with a "Why here?" breakdown.

Understand this area
Five everyday questions answered from the model's own numbers.

Rain on real cities
Google Photorealistic 3D with PaaniPulse overlays and rain, on a Maps Demo Key.

100 mm on 9 km² of Madurai: where does it go?



Modelled ponding depth, hypothetical 100 mm · 99 × 99 cells of about 30 m

Simulated · hypothetical scenario

5,90,412 m³
runoff generated; mean coefficient 0.644 from land cover

5,26,187 m³
held in terrain hollows (89%); 64,224 m³ leaves the area

382 / 125
hollows reached / overflowing into the next

295 hotspots
28 higher, 267 moderate relative potential

More rain, more water held

50 mm	2,70,095 m³ · 236 hotspots
100 mm	5,26,187 m³ · 295 hotspots
150 mm	7,72,378 m³ · 314

Why hotspot 1 ranks first (5 of 7 points, "higher")

- 2 pt — a hollow that holds 1.28 m of water
- 1 pt — almost flat, 0.8% slope
- 2 pt — 0.0 m above the nearest drainage line
- 0 pt — runoff collects here rather than passing through

RELIABILITY AND LIMITATIONS

Built to be clear about what it knows.

How we keep it honest `in the app`

- 123 automated tests, including analytic checks: a pit's volume within 1%, exact HAND in a V-valley, water conserved through fill-and-spill.
- Every modelled output is labelled simulated; categories are relative, never probabilities.
- Hotspots on hollows deeper than 3 m or on permanent water lose confidence and say why.
- Documented news reports are shown as context only, never as validation.
- Fallbacks: alternative elevation source on request; the 2D map if 3D or Google fails.

What the model does not do `limits`

- About 30 m elevation: kerbs, drains and narrow channels are invisible.
- Surface model: buildings, walls and bridges can create false hollows.
- Runoff coefficients are documented assumptions, not calibrated.
- No drainage infrastructure and no real-time rainfall input.
- Rain is uniform with no duration: a "what if", not a flood forecast.
- Only the selected area is modelled; results depend on it and on data coverage.

IMPACT AND FUTURE SCOPE

Making water flow easy to understand.

Readable

Turns elevation and flow analysis into maps and everyday sentences.

Exploratory

Shows where water may collect under a chosen "what if" rainfall.

Visual

Lets people see rain meet a real landscape, in 2D and 3D.

Open and free

Open data, no install, no account; runs on any modern browser.

Working today `built`

- Any area in India up to 25 km², from open global data
- Terrain analysis, rainfall scenarios and ranked hotspots
- 2D map, 3D terrain and plain-language explanations
- Documented Madurai reports shown alongside, as context

Possible next steps `not built`

- Drainage network data, where cities publish it
- Observed or forecast rainfall inputs, with intensity and duration
- Calibration against recorded waterlogging events
- Finer terrain models and systematic validation

PaaniPulse

Understand the terrain.
Understand the water.

See where rainwater goes.

LIVE
programmer-mukeshkrishnaank.github.io/pannipulse

CODE
github.com/Programmer-MukeshKrishnaaNK/pannipulse

HackCult '26

